

Chapter 2: Plants: Structure and Function

Textbook Exercise Solutions & Answers



1. Give examples of 3 plants that have:

(a) Spiny fruits

Jack fruit, Pineapple, Lychee (litchi), Datura

(b) Spiny stem

Cactus, Wild rose, Catclaw acacia, Silk, Cotton

(c) Red flowers

Rose, Dahlia, Hibiscus, Tulips,

(d) Yellow flowers

Marigold, Daffodil, Sunflower, Daisy

(e) Leaves which close at night

Mimosa plant, Tallwood, Prayer plant, Gulmohar

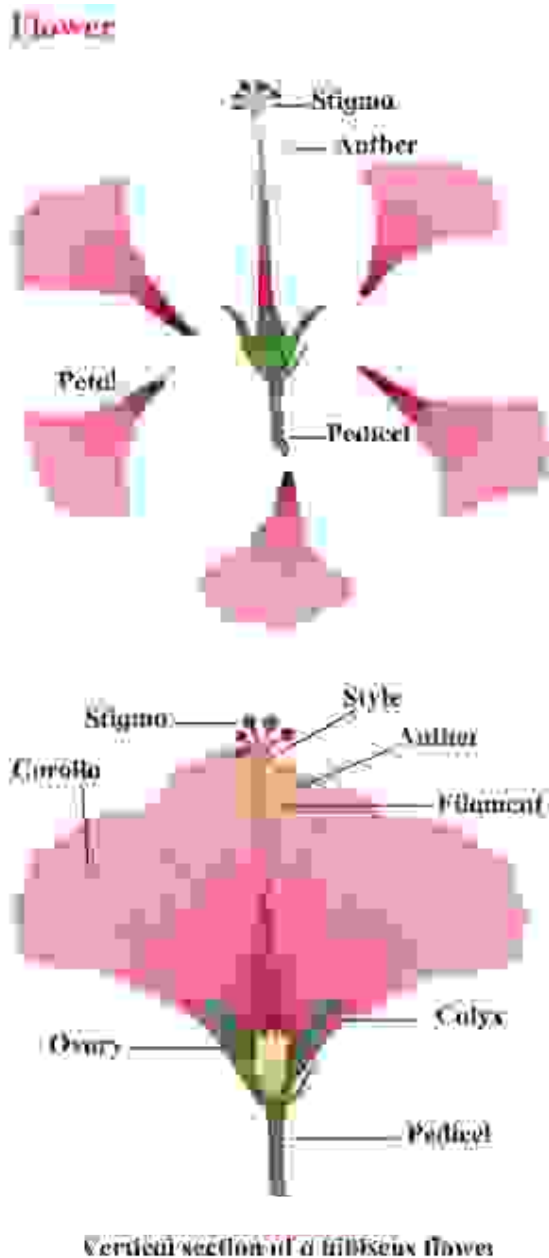
(f) Single seeded fruits

Mango, Lychee, Jamun

(g) Many seeded fruits

Custard apple, Papaya, Watermelon.

2. Observe any one flower and its various parts and describe it in your own words.



Vertical section of a tubiflorous flower

A flower has following parts:

1. Pedicel: (a) Flower may have a long or a short stalk called pedicel, (b) One end of the pedicel is attached to the stem, (c) The other end of the pedicel is expanded and swollen. It is called the receptacle, (d) Petals and other parts of the flower are supported on the receptacle.
2. Calyx: In the bud condition, the petals are covered by leaf like parts called sepals which are green in colour. They form the calyx.
3. Corolla: This is made up of colourful parts called petals.
4. Androecium: (a) This is the male reproductive part of the flower, (b) It consists of stamens, (c) Each stamen is made up of anther and filament.
5. Gynoecium: (a) This is the female reproductive part of the flower, (b) This is made up of carpels, (c) A carpel consists of stigma, style and ovary.

3. What are the similarities and differences between?

(a) Jowar and Moong

Differences:

Jowar	Moong
i. It has fibrous root.	i. It has tap root.
ii. It is a monocotyledonous seed.	ii. It is a dicotyledonous seed.
iii. It is a rabi crop.	iii. It is a kharif crop

Similarities:

Jowar and Moong both are angiosperms.

(b) Onion and Coriander

Differences:

Onion	Coriander
i. It is a biennial plant.	i. It is an annual plant.
ii. It has monocotyledonous seed.	ii. It has dicotyledonous seeds.
iii. It has fibrous root.	iii. It has tap root but later on changes to fibrous root system.

Similarities:

1. Both are used in cooking.
2. Both are edible.
3. Both belongs to kingdom plantae.

(c) Leaves of banana and Leaves of mango

Leaves of banana	Leaves of mango
i. It has parallel venation.	i. It has reticulate venation.
ii. It is very large in size.	ii. It is small in size.
iii. It is a monocotyledonous plant.	iii. It is a dicotyledonous plant.

(d) Coconut tree and Jowar stalk plant

Differences:

Coconut tree	Jowar stalk plant
i. It is tall and has a thick stem	i. It is small and has a thin stem.
ii. It has strong root system.	ii. It has weak root system.
iii. Each and every part of tree is useful.	iii. Only seeds are useful.

Similarities:

1. Both are from same kingdom plantae.
2. Both are autotrophic.
3. Both are monocotyledonous plants.

4. Explain the following images in your own words.

1. Diagram A is of maize seed. It is a monocotyledonous seed and does not divide into two equal parts. The plant has fibrous root system.
2. Diagram B is of bean seed. It is a dicotyledonous seed and it divides into two equal parts. The plant has tap root system.

5. Describe the functions of various parts of a plant.

The functions of various parts of a plant are as below.

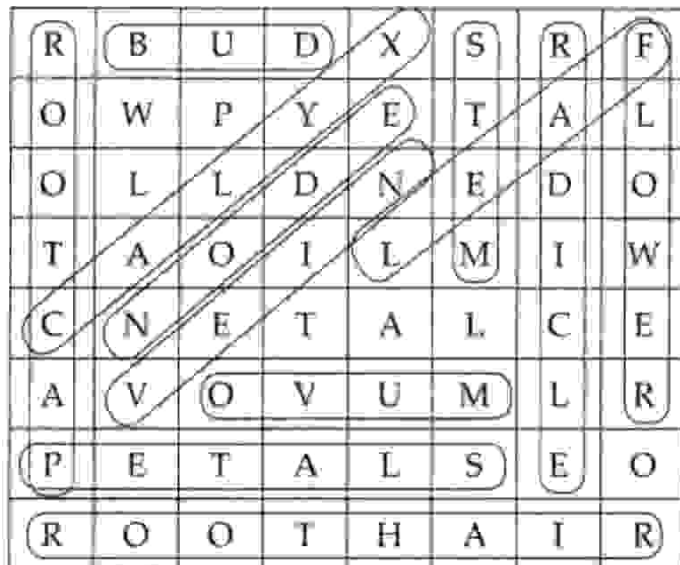
1. Root: (a) Fixation: It anchors the plant body to the soil, so helps in fixation, (b) Absorption: It absorbs water and nutrients from the soil, so helps in absorption, (c) Conduction: The root translocates water and mineral salts into the stem. (d) Storage: A certain amount of food is stored in the root which is utilized as it grows. (e) Preventing soil erosion: It helps to bind the soil particles and prevent them from being blown away by wind or water.
2. Stem: (a) It supports and holds leaves, flower and fruits, (b) The stem conducts the water and minerals from roots to leaves and fruits, (c) It stores the food.
3. Leaves: (a) It synthesizes food for the plant, (b) Stomata, the tiny openings in the leaf help in gaseous exchange and are responsible for the process of transpiration.
4. Flower: It helps in pollination.
5. Fruit: (a) It protects the seed, (b) It helps in seed dispersal.
6. Seed: A new plant develops from it.

6. Certain properties are mentioned below. Find a leaf corresponding to each property and describe those plants.

Certain properties are mentioned below. Find a leaf corresponding to each property and describe those plants. leaves with smooth surface, leaves with rough surface, fleshy leaf, spines on leaf.

1. Leaves with smooth surface: e.g. Banana leaf. It is large in size. It is closely rolled up one over the other. Together they look like a trunk but they form only an apparent trunk. It has parallel venation.
2. Leaves with rough surface: e.g. Hibiscus leaf. It has reticulate venation. Leaf margin is toothed.
3. Fleshy leaf: e.g. Jade plant, water hyacinth. It has fleshy, glossy and smooth leaves. They are coloured jade green and having a slightly red tinge towards the edge of leaves when exposed to higher level of light.
4. Spines on leaf: e.g. Opuntia, Ci/cas, kevda. Opuntia is a desert plant. Leaf is thick, fleshy and having spines on it.

7. Find the plant parts.



ROOT, ROOT CAP, ROOT HAIR, BUD, PETALS, STEM, RADICLE, FLOWER, LEAF, VEIN, CALYX, NODE, OVUM.